# Farming problems and the carbon cycle

*9B Plant growth · Lesson 6 of 7*

Date _____ Class _____

## Today I can 今天我能

- ☐ I can explain how fertilisers, pesticides and clearing land can harm wild organisms, including eutrophication and bioaccumulation.
- ☐ I can use the carbon cycle to explain how farming and burning fuels change the carbon dioxide in the air, and suggest ways to farm more sustainably.

## What happens today

1. Do Now: the dead fish
2. Learn: Eutrophication · Pesticide problems · A food web from a hedge next to a wheat field · and more
3. Core tasks
4. Challenge, then the exit ticket

Student Book pages 32 to 33 · Workbook pages 23 to 24

**What I learned today (one or two sentences) 今天我学到了：**

# Key Vocabulary

| Term | 中文 | Definition |
|---|---|---|
| **habitat** | 栖息地 | the place where an organism lives |
| **eutrophication** | 富营养化 | a very rich supply of nutrients in water, which leads to fast plant growth and then death of fish |
| **algae** | 藻类 | simple plant-like organisms that photosynthesise in water |
| **decomposer** | 分解者 | a microorganism that breaks down dead material, using oxygen |
| **persistent** | 持久性的 | does not break down in the environment |
| **bioaccumulation** | 生物富集 | the build-up of a substance in organisms along a food chain |
| **food web** | 食物网 | a diagram of linked food chains showing what eats what |
| **biodiversity** | 生物多样性 | the number of different species in an area |
| **carbon cycle** | 碳循环 | the processes that move carbon between the air, living things and the Earth |
| **combustion** | 燃烧 | burning, which releases carbon dioxide |
| **global warming** | 全球变暖 | the rise in the Earth's average temperature |
| **sustainable development** | 可持续发展 | making the things we need without destroying habitats for the future |

**Command words 指令词:** Draw 画出 · Explain 解释 · Match 连线（配对） · Name 说出名称 · Number 编号（排序） · Suggest 提出（建议） · Tick 打勾

# Quick Review

*the dead fish*

**1** What do fish need from the water for respiration?

**2** Name one farm chemical that rain can wash into a river or lake.

**3** Suggest how these fish died.

# Core 1 of 4

*Everyone does these. Answer on the card or under it. 人人都做，答案写在卡片上或下面。*

**1** Number the statements 1 to 6 to show how using fertiliser can lead to the death of fish. The first and last have been done for you.

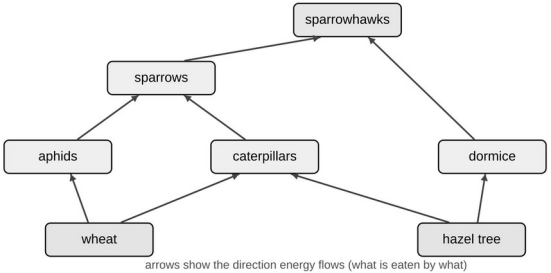
| Order | Statement |
|---|---|
| 1 | Nutrients from the fertiliser dissolve in soil water. |
| | Decomposers break down the dead plants, using oxygen from the water. |
| | Extra nutrients in the water cause rapid growth of algae and water plants. |
| | Soil water with extra nutrients leaks into a nearby stream or river. |
| | Algae and water plants at the surface block the light, so deeper plants die. |
| 6 | Fish die because there is too little oxygen in the water. |

# Core 2 of 4

*Everyone does these. Answer on the card or under it. 人人都做，答案写在卡片上或下面。*

**2** The diagram shows a food web from a hedge next to a wheat field.

**A food web from a hedge next to a wheat field**




arrows show the direction energy flows (what is eaten by what)

**a)** Name one producer in this food web.

# Core 3 of 4

*Everyone does these. Answer on the card or under it. 人人都做，答案写在卡片上或下面。*

**2** *continued*

**b)** Name two animals that eat caterpillars.

**c)** The farmer sprays the wheat with a general insecticide, which kills all insects. Tick one box in each row to show what happens to each population.

| Population | Increases | Decreases | No change |
|---|---|---|---|
| aphids | | | |
| caterpillars | | | |
| sparrows | | | |
| sparrowhawks | | | |

# Core 4 of 4

*Everyone does these. Answer on the card or under it. 人人都做，答案写在卡片上或下面。*

**3** Match each process in the carbon cycle to what it does. Draw a line.

| | |
|---|---|
| photosynthesis | releases carbon dioxide when fuels or wood burn |
| respiration | passes carbon compounds from plants to animals |
| combustion | removes carbon dioxide from the air |
| decomposers feeding | releases carbon dioxide from dead plants and animals |
| animals feeding | releases carbon dioxide from living cells, day and night |

# Challenge 1 of 2

*When Core is finished. Full sentences. 完成基础题后做；请写完整句子。*

**C1** In 19th century Ireland, most people survived by growing and eating one variety of potato, called the lumper. Between 1845 and 1852 more than a million people starved. Suggest why, using the words variety, disease and biodiversity.

# Challenge 2 of 2

*When Core is finished. Full sentences. 完成基础题后做；请写完整句子。*

**C2** A persistent insecticide was sprayed on crops next to a river for many years. Years later, fish-eating birds at the top of the river food chain were found to contain large amounts of the insecticide.

**a)** Explain how the insecticide reached the birds and why they contained so much of it.

**b)** In 2014 the EU banned some neonicotinoid insecticides on rapeseed, which flowers, but not on wheat. Suggest why.

# Exit Ticket

*Hand this card in at the end of the lesson.*

Name _____

**1** What does eutrophication mean?

**2** Name two processes that release carbon dioxide into the air.

**3** Why is it risky to plant one variety of crop over a very large area?

**Today I can:** Goal 1 ☐ Goal 2 ☐